PART B1

STRUCTURED QUESTIONS (70 MARKS)

INSTRUCTION:

1. Please answer all questions in the space provided within 2 hours 20 minutes (140 minutes);
2. For each submission, save and convert this file into pdf format using your matric card ID and section (matric-section.pdf), *e.g.* **A18CS8088-10.pdf**;
3. Upload the pdf file using the link provided in the e-learning.

QUESTION 1 [20 *Marks*]

A paging system with an *X*-bytes of logical memory with some portions of logical addresses (decimal) with its contents as shown in Table 1 need to be mapped to an *Y*-bytes of physical memory based on the complete 2^3 entries for a page table shown in Table 2. Assume that each entry size is 2^2-bytes and the physical memory size is 2^6 frames. Answer the following questions.

Table 1

Logical Address	Content
0	B
1	-
2	-
3	A
4	D
5	G
. . .	
19	E
20	-
21	H
22	C
23	F
. . .	

Table 2

Page#	Frame#
0	20
1	45
2	5
3	12
4	10
5	9
6	30
7	4

(a) Calculate the size of *X* and *Y*. Show your working. [4 *Marks*]

Answer:

→ *each entry in page table = page size = frame size = 4 bytes*

→ *the page table (Table 2) consists of 8 rows mean 8 pages for logical memory.*

→ *total frames for physical memory =* 2^6 *= 64 frames*

Logical memory:

*X = total pages * page size = 8 * 4 = 32 bytes*

Physical memory:

*Y = total frames * frame size = 64 * 4 = 256 bytes*

(b) What will be the contents of frame 9? Justify your answer. [3 *Marks*]

Answer:

→ *From Table 2, frame 9 is for page 5*

→ *Page 5 addresses are at 20, 21, 22, 23*

→ *Those addresses consists of {-, H, C, F}*

(c) How many bits will be used for the page number and frame number in logical memory address and physical memory address, respectively? Show your working. [4 *Marks*]

Answer:

→ *offset bit = bit for page size or frame size = 2*

logical memory: → $X = 32 = 2^5$	*physical memory:* → $Y = 256 = 2^8$
page number bit *= logical memory bit - page offset bit* *= 5 – 2 = 3 bits*	*page number bit* *= logical memory bit - page offset bit* *= 8 – 2 = 6 bits*

(d)	Calculate the logical address (decimal) given a physical address as 10110110_2. Show your working. [4 *Marks*] *Answer:* *Physical address → frame# : frame offset → 101101 : 10 → 45 : 2* *Logical address = (page# * page size) + offset* *= (1 * 4) + 2* *= 5 + 2 = 7*
(e)	Calculate the physical address of data ***E*** in binary of 8 bits. Assume that the initial offset is 0. Show your working. [3 *Marks*] *Answer:* *→ Data E in page 4, offset 3, frame 10* *Physical address E* *= (frame# * frame size) + offset E* *= (10 * 4) + 3* *= 40 + 3 = 43 =* 00101011_2
(e)	Write TWO consequences if the paging system resize the frame to 2^3-bytes? Justify each of your answer. [2 *Marks*] *Answer:* • *The size of page become larger to 8 bytes; because size of page = frame;* • *The frame number reduced to 32 frames; because (256 bytes / 8 bytes) = 32 frames;* • *The number reduced half to 4 pages; because (32 bytes / 8 bytes) = 4 pages;* • *The page table became smaller with only 4 entries (4 rows);because the entries = page number;*

QUESTION 2 [15 Marks]

(a) Consider a memory contains page of 100 bytes each and the addresses space as shown in Figure 1.

Page	Start address	End address
Page 0	0000	0100
Page 1	0100	0200
Page 2	0200	0300
Page 3	0300	0400
Page 4	0400	0500
Page 5	0500	0600
Page 6	0600	0700

Figure 1

→	0100	0321	0555	0111
	0400	0350	0100	0280
	0050	0050	0010	0080
	0603	0509	0455	0300
	0030	0100	0200	0330
	0010	0520	0100	0001

Figure 2

i) If CPU accesses the addresses shown in Figure 2, write the complete reference string. (Note: The arrow indicates the starting address from the first row). [2 *Marks*]

Ans:

Reference string: ***0 3 5 1*** *4 3 1 2* ***0 0 0 0*** *6 5 4 3* ***0 1 2 3*** *0 5 1 0*

ii) If the system using any policy of the page replacement with 7 frames for the reference string above, what will be the consequence to the page faults? Briefly justify your answer. [3 *Marks*]

Ans:

- *There will be maximum 7 page faults*
- *Because all pages are already in the frames*
- *No need to replace any frame*

(b) Table 3 illustrates the frames used with the current page reference using LRU page replacement algorithms at time t_{10}. Answer the following questions.

Table 3

Frame	0	1	2	3
Page Reference	2	1	3	4

i) Consider the following new page reference string:

2 1 5 6 2 1 2 3 7 6 3 2 1 2 3 5

By constructing a table with the page reference in the frames, identify how many page faults would occur using the same page replacement algorithms? Show your working. [7 *Marks*]

Ans:

4 frames: 7 page faults

3	1	2	4	2	1	5	6	2	1	2	3	7	6	3	2	1	2	3	5
			2			1	1				1	1	6			6			5
			1			2	2				2	2	2			2			2
			3			5	5				3	3	3			3			3
			4			4	6				6	7	7			1			1

- [1m] clear table with reference string and 4 frames
- [1m] correct columns with references (as shown in table)
- [4m] correct page replaced using LRU (page red in colors)
- [1m] indicate/state 7 page faults occurred

ii) Based on the previous answer, how to reduce the page fault number? Briefly, justify your answer. [3 *Marks*]

Ans:

- *Increase the frame number*
- *More frame number, more page reference will be in the frame,*
- *so reduce the page replacement / less page fault*

QUESTION 3 [8 *Marks*]

Consider a portion of directory structure for a workstation as illustrated in Figure 3. Assume that a command prompt console is using to perform some operations using the UNIX command line and the current working directory is *outdoor* folder. Answer the following questions in sequence.

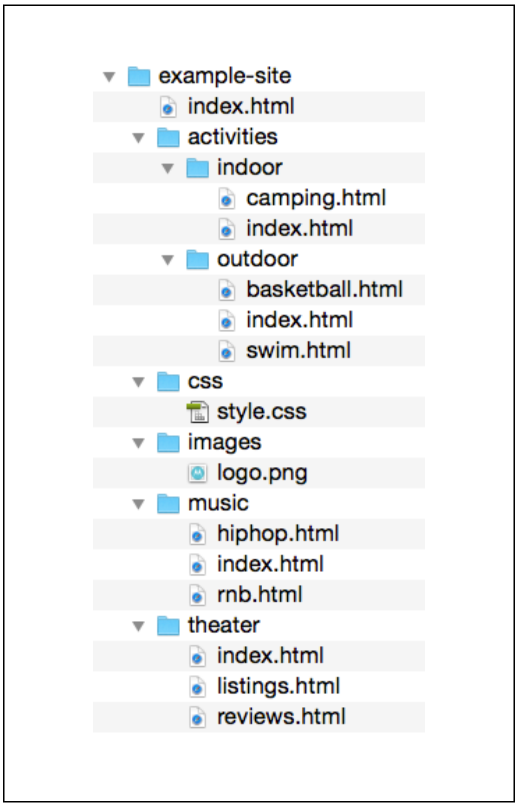


Figure 3

(*Source: https://stuyhsdesign.files.wordpress.com/2015/09/directory-structure1.png*)

(a) In the current directory, write a single command line to copy the *logo.png* into the *theater* folder. [3 *Marks*]

Answer:

```
cp ../../images/logo.png ../../theater
```

(b) In the current directory, write a single command line to change the working directory to *music* folder. [2 *Marks*]

Answer:

```
cd ../../music
```

(c) In the new current directory, write TWO different ways of single command line to set the access control of its files so that all classes can be read only. [3 *Marks*]

Answer:

- `chmod a+r,a-wx *.html`
- `chmod 444 *.html`
- `chmod a+r,a-wx hiphop.html index.html rnb.html`

QUESTION 4 [16 Marks]

Consider the following Figure 4 illustrates a secondary disk space with a file stored in some blocks sequentially. Answer the following questions.

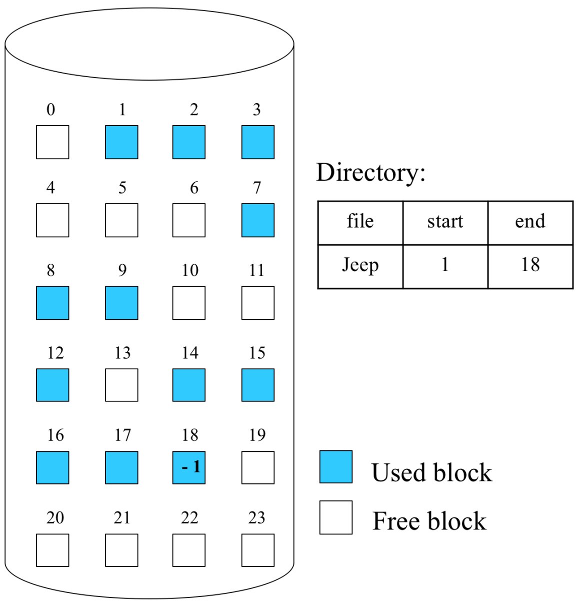


Figure 4

(a) Write the free space representation using the following methods clearly:

i) Grouping (assume each block can only store 3 block addresses) [2 *Marks*]

Answer:

Block 0	Block 6	Block 13	Block 21
4	10	19	22
5	11	20	23
6	13	21	-

	ii)	Counting [2 *Marks*] *Answer:* 0, 0 4, 2 10, 1 13, 0 19, 4
(b)	If the capacity of storage is 98304 bytes, and the number of blocks as shown in Figure 4, what is the size of file *jeep* in bytes? Assume all blocks fully occupied. Show your working. [2 *Marks*]. *Answer:* $block\ size = \frac{capacity}{no\ of\ blocks} = \frac{98304}{24} = 4096$ $jeep\ file\ size = no\ of\ blocks\ x\ 4096 = 12\ x\ 4096 = 49152\ bytes$	
(c)	Consider the file *jeep* is edited, and requires some block of storage to be appended at the end of the file.	
	i)	If the original size of the file *jeep* is 22.528 Kb that occupied all the blocks used as shown in Figure 4, how many additional blocks are required when the file size becomes 30.176 Kb? Show your working. [2 *Marks*] *Answer:* Original total size = 24.576 Kb Blocks used = 12 Size each block = 24576 / 12 = 2048 bytes → (30.176 - 24576) / 2048 = 5600 / 2048 = 2.73 blocks = 3 blocks or → (30176 / 2048) - 12 = 14.734 - 12 = 2.73 blocks = 3 blocks
	ii)	Based on the previous answer, state the block number that produced the internal fragmentation and calculate the size. Show your working. [2 *Marks*] *Answer:* → block 21 Internal fragmentation = 5600 – (2048*2) = 5600 – 4096 = 1504 Kb

(d) While checking for block consistency, the result of scanning 16 blocks of storage is shown in Figure 5.

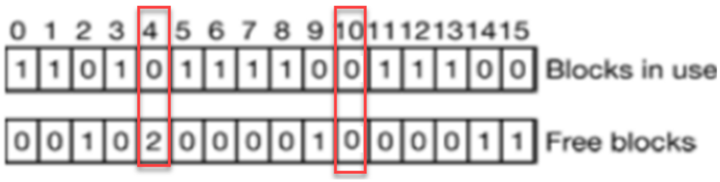
Block numbers

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Block in used	1	1	0	1	0	1	1	1	1	0	0	1	1	1	0	0
Free blocks	0	0	1	0	2	0	0	0	0	1	0	0	0	0	1	1

Figure 5

i) Based on the Figure 5, determine the consistency error(s) exist by stating to the block number of the storage. Justify your answer. [4 *Marks*]

Answer:



Two type of error

- [Duplicate block in free list] at block number 4 – number on free block shouldn't be 2 – it can either be 0 or 1
- [Missing block] at block number 10 – both free block and in use blocks have a value of 0 – either one should be 0, while the other one should be 1

ii) For the error(s) stated in previous answer, suggest a solution to overcome the error(s) [2 *Marks*]

Answer:

- At block 4 – it is not possible to have 2 reference of free block – solution rebuild the free list
- At block 10 – both have a value of 0, indicates that it is not used and not free – solution return that block to free blocks (free blocks 1, and block in use 0).

QUESTION 5 [11 *Marks*]

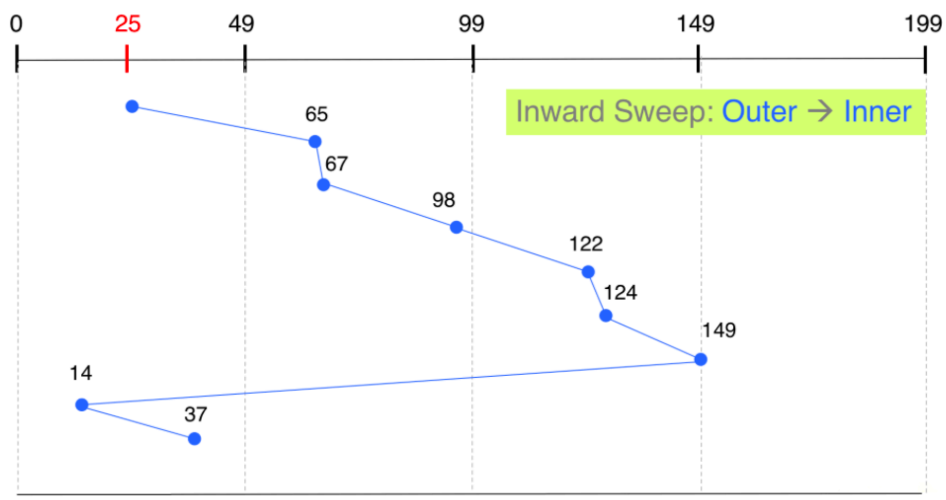
Consider the following information:

- ✓ A disk drive with 200 cylinders;
- ✓ A request queue :
  98, 149, 37, 122, 14, 124, 65, 67
- ✓ Head pointer 25.
- ✓ Direction: the disk arm is moving to innermost.

Answer the following questions.

(a) By using the C-LOOK disk scheduling algorithms, illustrates by drawing the movement of the disk arm to satisfy the requests. Label clearly. [7 *Marks*]

Ans:



- *General diagram with track (1m)*
- *Correct movement toward 199 (1m)*
- *Correct track visited (4m)*
- *Correct concept from 149 to 14 (1m)*

(b)	Calculate the total distance (in cylinders) that the disk arm moves to satisfy the requests. Show your working. [2 *Marks*] *Ans:* *= (149 – 25) + (149 – 14) + (37 – 14)* *= 124 + 135 + 23* *= 282 cylinders (head movement)*
(c)	If the LOOK disk scheduling algorithms is used, briefly describe a conclusion that can be made about the disk arm movement to satisfy the requests. [2 *Marks*] *Ans:* *= (149 – 25) + (149 – 14) = 124 + 147 = 271 cylinders (head movement)* → *LOOK uses 11 cylinders (head movement) less than C-LOOK*

------- End of Question -------

GOOD LUCK !